

Assignment 1

Java RMI Oral Questions

1. What is RMI?

RMI (Remote Method Invocation) is a Java technology that allows one Java program to call methods of an object located on another machine.

2. What is the full form of RMI?

Remote Method Invocation.

3. Why is RMI used?

RMI is used for distributed applications where client and server communicate remotely.

4. What is a remote object?

A remote object is an object whose methods can be invoked from another JVM or machine.

5. Which package is mainly used in RMI?

java.rmi

6. Which interface must a remote interface extend?

java.rmi.Remote

7. Why do we use RemoteException?

It handles communication errors during remote method calls.

8. What is a stub in RMI?

A stub acts as a proxy on the client side that forwards requests to the remote object.

9. What is a skeleton in RMI?

A skeleton receives requests from the stub and passes them to the actual remote object on the server.

10. What is UnicastRemoteObject?

It is a class used to create remote objects that can receive incoming remote calls.

11. What is the role of RMI Registry?

RMI Registry stores remote object references so clients can find remote services.

12. Which command is used to start RMI Registry?

rmiregistry

13. What is the purpose of Naming.rebind()?

It binds the remote object with a name in the RMI registry.

Example:

```
Naming.rebind("rmi://localhost/FactorialService", c);
```

14. What is the purpose of Naming.lookup()?

It searches and returns the remote object from the registry.

15. What is the function of the client program?

The client requests services from the remote server object.

16. What is the function of the server program?

The server creates and hosts remote objects.

17. What is the problem statement in this practical?

To create an RMI application for finding factorial of a number.

18. Why is BigInteger used in factorial program?

Because factorial values become very large and int cannot store them.

19. What are the steps to run an RMI application?

1. Compile programs using javac
2. Start rmiregistry

3. Run server
4. Run client

20. Which command is used to compile Java files?

javac filename.java

21. What is distributed computing?

Distributed computing means different computers work together over a network.

22. Difference between local method call and remote method call?

Local Call	Remote Call
Happens in same JVM	Happens on another machine
Faster	Slower
No network needed	Network required

23. What is the default port of RMI Registry?

1099

24. Which exceptions are handled in client program?

- MalformedURLException
 - RemoteException
 - NotBoundException
 - ArithmeticException
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25. What is the outcome of this practical?

Successfully implementing client-server communication using RMI.